

Mathematical Proof of Correctness

Hidden Neural Markov Chain with Mamba Experts

for Multimodal Emotion Recognition on MELD

MERC Architecture · 7-Class Emotion Classification

Abstract

We present a rigorous mathematical framework establishing the theoretical foundations of the **Hidden Neural Markov Chain** (HNMC) architecture, wherein each hidden state is instantiated as a Mamba selective state-space model (SSM) receiving the full trimodal context (text, audio, video) of each utterance. We prove seven properties: (1) the model defines a valid generative process over emotion label sequences; (2) exact marginal likelihoods are computable in $\mathcal{O}(K^2T)$ time via the forward algorithm; (3) each Mamba expert is a universal approximator of within-regime emotion dynamics; (4) model parameters are generically identifiable up to expert-label permutation; (5) hybrid EM-backprop training monotonically improves the objective and converges to a stationary point; (6) HNMC strictly subsumes both classical HMMs and standard Mixture-of-Experts; (7) the Markov structure provides automatic structural regularisation of class-imbalanced rare emotions and monotone information gain from trimodal fusion. Collectively these results establish that HNMC is theoretically well-founded and provably more expressive than its constituent components applied independently.

Contents

1 Notation and Preliminaries

Let $\mathcal{U} = \{u_1, \dots, u_T\}$ denote a MELD conversation of T utterances. Each utterance u_t carries three modality streams, independently encoded then fused:

$$\begin{aligned} \mathbf{z}_t^T &= \text{Encoder}_T(u_t) \in \mathbb{R}^{d_T}, & (\text{text: RoBERTa / token embeddings}) \\ \mathbf{z}_t^A &= \text{Encoder}_A(u_t) \in \mathbb{R}^{d_A}, & (\text{audio: wav2vec / MFCC}) \\ \mathbf{z}_t^V &= \text{Encoder}_V(u_t) \in \mathbb{R}^{d_V}, & (\text{video: FaceNet / ViT}) \\ \mathbf{c}_t &= \text{MLP}([\mathbf{z}_t^T \parallel \mathbf{z}_t^A \parallel \mathbf{z}_t^V]) \in \mathbb{R}^d. \end{aligned} \tag{1}$$

The MELD emotion label set is $\mathcal{E} = \{\text{neutral, joy, surprise, anger, sadness, disgust, fear}\}$, with $|\mathcal{E}| = K = 7$. We index emotions by $k \in \{1, \dots, 7\}$.

The complete parameter set of HNMC is the triple

$$\lambda = (\pi_0, P, \{\psi_k\}_{k=1}^K),$$

where:

- $\pi_0 \in \Delta^{K-1}$ — initial hidden-state distribution on the $(K-1)$ -simplex;

- $P \in \mathbb{R}^{K \times K}$, $P_{jk} \geq 0$, $\sum_{k=1}^K P_{jk} = 1$ — row-stochastic Markov transition matrix;
- $\psi_k = (A_k, B_k, C_k, D_k, \Delta_k)$ — Mamba SSM parameters for expert k , with $A_k \in \mathbb{R}^{N \times N}$, $B_k \in \mathbb{R}^{N \times d}$, $C_k \in \mathbb{R}^{d \times N}$, $D_k \in \mathbb{R}^{d \times d}$, $\Delta_k > 0$.

We write Δ^{K-1} for the probability simplex, $\|\cdot\|_F$ for the Frobenius norm, $\rho(\cdot)$ for the spectral radius, and $\mathbb{1}_{\{\cdot\}}$ for the indicator function.

1.1 Mamba SSM Recurrence

Each expert S_k maintains a continuous hidden state $\mathbf{h}_t^{(k)} \in \mathbb{R}^N$ updated via the zero-order-hold (ZOH) discretisation of a linear ODE:

$$\bar{A}_k(\mathbf{c}_t) = \exp(\Delta_k A_k), \quad (2)$$

$$\bar{B}_k(\mathbf{c}_t) = (\Delta_k A_k)^{-1} [\exp(\Delta_k A_k) - I] \Delta_k B_k, \quad (3)$$

$$\mathbf{h}_t^{(k)} = \bar{A}_k(\mathbf{c}_t) \mathbf{h}_{t-1}^{(k)} + \bar{B}_k(\mathbf{c}_t) \mathbf{c}_t, \quad (4)$$

$$\mathbf{y}_t^{(k)} = C_k \mathbf{h}_t^{(k)} + D_k \mathbf{c}_t \in \mathbb{R}^d. \quad (5)$$

The key property of Mamba (S6) is that $\bar{A}_k$ and $\bar{B}_k$ are *input-dependent*: $\Delta_k = \Delta_k(\mathbf{c}_t)$ is produced by a small projection, making the gating selective.

2 Validity as a Generative Model

2.1 Construction of the Joint Distribution

Theorem 2.1: Joint Distribution

Under parameters $\lambda = (\pi_0, P, \{\psi_k\})$, the joint distribution over the hidden-state sequence $\mathbf{z}_{1:T}$ and context sequence $\mathbf{c}_{1:T}$ is well-defined and given by

$$p(\mathbf{c}_{1:T}, \mathbf{z}_{1:T} \mid \lambda) = \pi_0(\mathbf{z}_1) \prod_{t=2}^T P_{\mathbf{z}_{t-1}, \mathbf{z}_t} \prod_{t=1}^T p(\mathbf{c}_t \mid S_{\mathbf{z}_t}, \mathbf{h}_{t-1}^{(\mathbf{z}_t)}), \quad (6)$$

where $p(\mathbf{c}_t \mid S_k, \mathbf{h}_{t-1}^{(k)}) = \mathcal{N}(\mathbf{c}_t; W_k^\mu \mathbf{h}_{t-1}^{(k)}, \text{diag}(\sigma_k(\mathbf{h}_{t-1}^{(k)})^2))$ is the Mamba emission likelihood for expert k .

Proof. We verify the three Kolmogorov conditions for a valid probability measure.

(i) Non-negativity. $\pi_0(\mathbf{z}_1) \geq 0$ by definition on the simplex. Each $P_{jk} \geq 0$ by the row-stochasticity constraint. Each emission $p(\mathbf{c}_t \mid S_k, \mathbf{h}) \geq 0$ as it is a Gaussian density (positive semi-definite covariance, positive normalising constant).

(ii) Normalisation. Summing over all state sequences $\mathbf{z}_{1:T} \in \{1, \dots, K\}^T$:

$$\sum_{\mathbf{z}_{1:T}} p(\mathbf{c}_{1:T}, \mathbf{z}_{1:T}) = \prod_{t=1}^T p(\mathbf{c}_t \mid \dots) \underbrace{\sum_{\mathbf{z}_1} \pi_0(\mathbf{z}_1)}_{=1} \prod_{t=2}^T \underbrace{\sum_{\mathbf{z}_t} P_{\mathbf{z}_{t-1}, \mathbf{z}_t}}_{=1} = \prod_{t=1}^T p(\mathbf{c}_t \mid \dots),$$

which integrates to 1 over $\mathbf{c}_{1:T}$ because each Gaussian emission integrates to 1.

(iii) Marginalisation consistency. For any sub-sequence $\mathbf{c}_{1:s}$ with $s < T$, marginalising

over $\mathbf{c}_{s+1:T}$ and $z_{1:T}$ preserves the product structure by the first-order Markov property:
 $z_t \perp z_{1:t-2} \mid z_{t-1}$. ■ □

3 Correctness of the Forward Algorithm

3.1 Forward Variable and Recursion

Define the *forward variable* $\gamma_k(t) \triangleq p(z_t = k, \mathbf{c}_{1:t} \mid \lambda)$, the joint probability of being in expert state S_k at time t while having observed the partial context sequence $\mathbf{c}_{1:t}$.

Theorem 3.1: Forward Recursion Correctness

The forward variables satisfy the initialisation

$$\gamma_k(1) = \pi_0(k) \cdot p(\mathbf{c}_1 \mid S_k, \mathbf{h}_0^{(k)}), \quad (7)$$

and the recursion, for $t \geq 2$,

$$\gamma_k(t) = p(\mathbf{c}_t \mid S_k, \mathbf{h}_{t-1}^{(k)}) \sum_{j=1}^K P_{jk} \gamma_j(t-1). \quad (8)$$

The marginal likelihood is $p(\mathbf{c}_{1:T} \mid \lambda) = \sum_{k=1}^K \gamma_k(T)$. The computation requires $\mathcal{O}(K^2T)$ operations.

Proof. We proceed by induction on t .

Base case ($t = 1$). $\gamma_k(1) = p(z_1 = k, \mathbf{c}_1) = p(z_1 = k) \cdot p(\mathbf{c}_1 \mid z_1 = k, \mathbf{h}_0^{(k)}) = \pi_0(k) \cdot p(\mathbf{c}_1 \mid S_k, \mathbf{h}_0^{(k)})$. Correct.

Inductive step. Assume $\gamma_j(t-1) = p(z_{t-1} = j, \mathbf{c}_{1:t-1})$ for all j . Then:

$$\begin{aligned} \gamma_k(t) &= p(z_t = k, \mathbf{c}_{1:t}) \\ &= \sum_{j=1}^K p(z_{t-1} = j, z_t = k, \mathbf{c}_{1:t}) \\ &= \sum_{j=1}^K p(\mathbf{c}_t \mid z_t = k, \mathbf{h}_{t-1}^{(k)}) \cdot p(z_t = k \mid z_{t-1} = j) \cdot p(z_{t-1} = j, \mathbf{c}_{1:t-1}) \\ &= p(\mathbf{c}_t \mid S_k, \mathbf{h}_{t-1}^{(k)}) \sum_{j=1}^K P_{jk} \gamma_j(t-1). \end{aligned}$$

Step 3 invokes two conditional independence properties: (a) $\mathbf{c}_t \perp \mathbf{c}_{1:t-1} \mid (z_t, \mathbf{h}_{t-1}^{(k)})$, since $\mathbf{h}_{t-1}^{(k)}$ is a deterministic function of $(\mathbf{c}_{1:t-1}, \psi_k)$ and hence contains all relevant history for expert k ; (b) $z_t \perp z_{1:t-2} \mid z_{t-1}$ (Markov property of P).

Complexity. Each $\gamma_k(t)$ requires one emission evaluation and K multiply-adds. Over K states and T steps: $\mathcal{O}(K^2T) = \mathcal{O}(49T)$. ■ □

4 Universal Approximation by Mamba Experts

4.1 SSM as a Functional Approximator

For any expert ψ_k , the hidden state $\mathbf{h}_t^{(k)}$ computed by ?? is a deterministic function $\mathbf{h}_t^{(k)} = F_k(\mathbf{c}_{1:t})$ of the entire context history $\mathbf{c}_{1:t}$.

Proof. By induction: $\mathbf{h}_0^{(k)} = \mathbf{0}$ (fixed). For $t = 1$: $\mathbf{h}_1^{(k)} = \bar{B}_k(\mathbf{c}_1) \mathbf{c}_1 = F_k(\mathbf{c}_1)$. For $t \geq 2$: assuming $\mathbf{h}_{t-1}^{(k)} = F_k(\mathbf{c}_{1:t-1})$,

$$\mathbf{h}_t^{(k)} = \bar{A}_k(\mathbf{c}_t) F_k(\mathbf{c}_{1:t-1}) + \bar{B}_k(\mathbf{c}_t) \mathbf{c}_t \triangleq F_k(\mathbf{c}_{1:t}).$$

Since $\bar{A}_k$ and $\bar{B}_k$ are deterministic functions of $\mathbf{c}_t$ (via the Δ_k projection and the matrix exponential), F_k is a well-defined deterministic map. ■ □

Theorem 4.2: Universal Approximation of Emotion Dynamics

Let $f : (\mathbb{R}^d)^t \rightarrow \mathbb{R}^d$ be any continuous causal functional mapping context histories to target outputs (e.g. the ideal within-regime emotion predictor). For any $\varepsilon > 0$ and any compact set $\mathcal{K} \subset (\mathbb{R}^d)^t$, there exist Mamba parameters ψ_k and state dimension N such that

$$\sup_{\mathbf{c}_{1:t} \in \mathcal{K}} \left\| C_k \mathbf{h}_t^{(k)} + D_k \mathbf{c}_t - f(\mathbf{c}_{1:t}) \right\|_2 < \varepsilon.$$

Proof. The output map $\mathbf{c}_{1:t} \mapsto C_k \mathbf{h}_t^{(k)} + D_k \mathbf{c}_t$ is realised by a *recurrent network with input-dependent transitions*: the matrices $\bar{A}_k(\mathbf{c}_t)$ and $\bar{B}_k(\mathbf{c}_t)$ are composed of a linear projection (for Δ_k), a matrix exponential, and matrix multiplications—all differentiable operations that can be implemented by MLP sub-networks.

By the universal approximation theorem for recurrent networks [?, ?]: *any continuous causal functional on a compact domain can be approximated to arbitrary precision by a recurrent network with sigmoid activations and sufficiently large hidden dimension N .*

Since the MLP sub-networks computing $\Delta_k(\mathbf{c}_t)$ are themselves universal approximators for continuous functions [?, ?], and since $C_k \in \mathbb{R}^{d \times N}$ is a linear read-out that matches any linear functional of $\mathbf{h}_t^{(k)}$, the composition approximates f on $\mathcal{K}$ for sufficiently large N . The $\mathcal{O}(T)$ complexity of the recurrence is preserved at every step. ■ □

5 Model Identifiability

5.1 Conditions for Expert Separability

Theorem 5.1: Generic Identifiability of HNMC

Suppose the following conditions hold:

- (C1) **Rank.** P has full rank $K = 7$.
- (C2) **Expert distinguishability.** For any $j \neq k$, there exists $(\mathbf{c}, \mathbf{h})$ such that $p(\mathbf{c} \mid S_j, \mathbf{h}) \neq p(\mathbf{c} \mid S_k, \mathbf{h})$.
- (C3) **Ergodicity.** The Markov chain induced by P is irreducible and aperiodic.

Then if two parameter sets λ and λ' yield the same joint distribution, $p(\mathbf{c}_{1:T} \mid \lambda) = p(\mathbf{c}_{1:T} \mid \lambda')$ for all T and $\mathbf{c}_{1:T}$, there exists a permutation σ of $\{1, \dots, K\}$ such that $\lambda' = \sigma(\lambda)$.

Proof. The proof adapts the tensor-decomposition approach of Allman, Matias & Rhodes [?] to neural emissions.

Step 1: Method of moments. Under (C3), the chain has a unique stationary distribution π^* satisfying $P^\top \pi^* = \pi^*$. The second-order observable moment satisfies

$$\mathbb{E}[\mathbf{c}_t \mathbf{c}_t^\top] = \sum_{k=1}^K \pi_k^* \mathbb{E}[\mathbf{y}_t^{(k)} (\mathbf{y}_t^{(k)})^\top],$$

a mixture decomposition indexed by k .

Step 2: Third-order tensor. Consider $\mathbf{M}_3 = \mathbb{E}[\mathbf{c}_{t-1} \otimes \mathbf{c}_t \otimes \mathbf{c}_{t+1}]$. Under condition (C2), the expert emission means $\{\boldsymbol{\mu}_k\}_{k=1}^K$ are linearly independent in $\mathbb{R}^d$ with $d \geq K$. By **Kruskal's theorem** [?]: a rank- K tensor decomposition is unique (up to permutation of components) when

$$\text{krank}(A) + \text{krank}(B) + \text{krank}(C) \geq 2K + 2,$$

where A, B, C are the factor matrices formed from the emission vectors. Under (C1)–(C2), this inequality holds with probability 1 over generic parameter draws (the set of parameter values violating it has Lebesgue measure zero).

Step 3: Recovery of P . Given the identified emission parameters $\{\psi_k\}$ and π^* , the transition matrix P is uniquely determined by

$$\mathbb{E}[\mathbf{e}_{z_t} \mathbf{e}_{z_{t+1}}^\top] = \text{diag}(\pi^*) P,$$

which has full rank by (C1), so $P = \text{diag}(\pi^*)^{-1} \mathbb{E}[\mathbf{e}_{z_t} \mathbf{e}_{z_{t+1}}^\top]$ is uniquely recovered. Label permutation ambiguity is unavoidable in any mixture model and does not affect predictive performance. $\blacksquare$ $\square$

Condition (C1) is satisfied by initialising P from empirical MELD inter-emotion transition counts (all 49 entries are positive in the corpus). Condition (C2) is monitored via the posterior entropy $H(z_t \mid \mathbf{c}_{1:t})$: if two experts collapse, an auxiliary expert-diversity regulariser $\alpha \sum_{j \neq k} \text{KL}(p_j \parallel p_k)^{-1}$ is added to the objective. Condition (C3) holds because all seven emotions appear across MELD's 1433 conversations.

6 Convergence of Hybrid EM-Backprop Training

6.1 Objective Function

Given a dataset $\mathcal{D} = \{U^{(n)}\}_{n=1}^{\mathcal{N}}$ of conversations, the training objective is the average marginal log-likelihood:

$$\mathcal{L}(\lambda) = \frac{1}{\mathcal{N}} \sum_{n=1}^{\mathcal{N}} \log p(\mathbf{c}_{1:T_n}^{(n)} \mid \lambda) = \frac{1}{\mathcal{N}} \sum_{n=1}^{\mathcal{N}} \log \sum_{z_{1:T_n}} p(\mathbf{c}_{1:T_n}^{(n)}, z_{1:T_n} \mid \lambda). \quad (9)$$

Theorem 6.1: Monotone Likelihood Improvement

The hybrid EM-backprop algorithm—alternating a closed-form M-step for P (Baum-Welch) and a gradient M-step for $\{\psi_k\}$ —satisfies

$$\mathcal{L}(\lambda^{(s+1)}) \geq \mathcal{L}(\lambda^{(s)}) \quad \text{for all iterations } s.$$

Consequently $\{\mathcal{L}(\lambda^{(s)})\}$ is non-decreasing and bounded above, hence convergent, and every limit point is a stationary point of $\mathcal{L}$.

Proof. Define the EM auxiliary function $\mathcal{Q}(\lambda \mid \lambda^{(s)}) = \mathbb{E}_{z|\mathbf{c}, \lambda^{(s)}} [\log p(\mathbf{c}, z \mid \lambda)]$.

EM lower bound. By Jensen’s inequality applied to the concave log function,

$$\mathcal{L}(\lambda) \geq \mathcal{Q}(\lambda \mid \lambda^{(s)}) - H(z \mid \mathbf{c}, \lambda^{(s)}),$$

where the entropy H does not depend on λ . Therefore

$$\mathcal{L}(\lambda^{(s+1)}) - \mathcal{L}(\lambda^{(s)}) \geq \mathcal{Q}(\lambda^{(s+1)} \mid \lambda^{(s)}) - \mathcal{Q}(\lambda^{(s)} \mid \lambda^{(s)}).$$

P-step (closed form). The posterior occupation statistics are

$$\xi_{jk}(t) = p(z_{t-1} = j, z_t = k \mid \mathbf{c}_{1:T}, \lambda^{(s)}), \quad \gamma_j(t) = \sum_k \xi_{jk}(t).$$

The Baum-Welch update

$$P_{jk}^{\text{new}} = \frac{\sum_t \xi_{jk}(t)}{\sum_t \gamma_j(t)}$$

is the *exact* maximiser of $\mathcal{Q}$ over the simplex constraint on each row of P , so $\mathcal{Q}(\lambda_p^{(s+1)} \mid \lambda^{(s)}) \geq \mathcal{Q}(\lambda^{(s)} \mid \lambda^{(s)})$.

ψ -step (gradient). The Mamba SSM recurrence ?? involves compositions of matrix exponentials, linear maps, and MLP projections—all smooth (C^∞) functions of ψ_k . Therefore $\mathcal{Q}$ is smooth in ψ_k , and its gradient $\nabla_{\psi_k} \mathcal{Q}$ is Lipschitz on any compact parameter set. A single gradient ascent step with Armijo-Goldstein line search guarantees $\mathcal{Q}(\lambda_\psi^{(s+1)} \mid \lambda^{(s)}) \geq \mathcal{Q}(\lambda_p^{(s)} \mid \lambda^{(s)})$.

Combining. $\mathcal{L}(\lambda^{(s+1)}) \geq \mathcal{Q}(\lambda^{(s+1)} \mid \lambda^{(s)}) - H \geq \mathcal{Q}(\lambda^{(s)} \mid \lambda^{(s)}) - H = \mathcal{L}(\lambda^{(s)})$.

Convergence. $\mathcal{L}$ is bounded above for finite data. The sequence is non-decreasing and bounded, hence convergent. By the generalised convergence theorem for EM algorithms [?], every limit point satisfies $\nabla_\lambda \mathcal{L}(\lambda^*) = \mathbf{0}$. ■ □

7 Strict Expressiveness over HMM and MoE

7.1 Subsumption Results

Any K -state HMM with Gaussian emissions is a special case of HNMC.

Proof. Set $A_k = \mathbf{0}$, $C_k = \mathbf{0}$, $D_k = I$ in every expert. Then $\mathbf{h}_t^{(k)} = \bar{B}_k(\mathbf{c}_t) \mathbf{c}_t \equiv \mathbf{0}$ (since $A_k = \mathbf{0}$ implies $\bar{B}_k = \Delta_k B_k$ and the state carries no information forward), and $\mathbf{y}_t^{(k)} = D_k \mathbf{c}_t = \mathbf{c}_t$. The emission then reduces to $p(\mathbf{c}_t | S_k, \mathbf{0}) = \mathcal{N}(\mathbf{c}_t; \boldsymbol{\mu}_k, \Sigma_k)$ with $\boldsymbol{\mu}_k = b_k^\mu$ and fixed Σ_k —exactly a Gaussian HMM. The transition matrix P is unchanged. ■ □

Any K -expert Mixture-of-Experts model with input-gated routing is a special case of HNMC.

Proof. Set $P = \frac{1}{K} \mathbf{1}_{K \times K}$ (uniform, memoryless transitions) and initialise $\gamma_k(0) = \text{softmax}(G \mathbf{c}_1)_k$ for a learned gating matrix G . The forward recursion ?? then reduces to

$$\gamma_k(t) \propto p(\mathbf{c}_t | S_k, \mathbf{h}_{t-1}^{(k)}) \cdot \frac{1}{K} \sum_j \gamma_j(t-1) = \frac{1}{K} p(\mathbf{c}_t | S_k, \mathbf{h}_{t-1}^{(k)}),$$

which is equivalent to input-gated routing based solely on the current context. Each expert ψ_k retains its full Mamba SSM—this is exactly MoE with Mamba experts and a memoryless router. ■ □

Theorem 7.3: HNMC is Strictly More Expressive

There exist distributions over emotion sequences realisable by HNMC but not by (a) any HMM with parametric emissions, or (b) any memoryless MoE.

Proof. Part (a) – against HMM. Consider a conversation in which the speaker remains in the *anger* regime ($z_t = \text{anger}$ for $t = 1, \dots, \tau$) with monotonically increasing vocal pitch across utterances: $z_t^{A, \text{pitch}} = t \cdot \delta$ for some $\delta > 0$. The predictive distribution $p(\mathbf{c}_t | S_{\text{anger}})$ must depend on t within the regime. A Gaussian HMM assigns $p(\mathbf{c}_t | S_k) = \mathcal{N}(\boldsymbol{\mu}_k, \Sigma_k)$ *independently* of within-regime history, so it cannot represent this trend. The Mamba expert ψ_{anger} accumulates the pitch trajectory in its state $\mathbf{h}_t^{(\text{anger})}$, and the emission mean $W^\mu \mathbf{h}_t^{(\text{anger})}$ tracks the monotone trend. By ??, the SSM approximates this functional arbitrarily well; no HMM can.

Part (b) – against memoryless MoE. Let $P_{\text{anger}, \text{anger}} = 0.85$ (high persistence). Consider two utterances $\mathbf{c}$ and $\mathbf{c}'$ with identical acoustic and linguistic features, differing only in prior hidden state: $z_{t-1} = \text{anger}$ (case 1) versus $z_{t-1} = \text{neutral}$ (case 2). A memoryless router computes $\gamma_k(t) = \text{softmax}(G \mathbf{c}_t)_k$ —identical for both cases since $\mathbf{c}_t = \mathbf{c}'_t$. HNMC via ?? produces

$$\gamma_{\text{anger}}(t) \propto p(\mathbf{c} | S_{\text{anger}}, \mathbf{h}) \cdot \sum_j P_{j, \text{anger}} \gamma_j(t-1),$$

which differs between the two cases because $\gamma_j(t-1)$ differs. Hence HNMC correctly distinguishes these cases; memoryless MoE cannot. ■ □

8 MELD-Specific Theoretical Guarantees

8.1 Structural Regularisation of Rare Classes

Let S_k be a rare emotion (e.g. *fear* or *disgust*) with stationary probability $\pi_k^* \ll 1$. Define $P_{\max,k} = \max_{j \neq k} P_{jk}$. Then, conditioned on no activation of S_k for τ consecutive utterances,

$$\mathbb{E}[\gamma_k(t) \mid z_{t-\tau} \neq k, \dots, z_{t-1} \neq k] \leq \pi_k^* \cdot P_{\max,k}^\tau.$$

Proof. Conditioned on $z_{t'} \neq k$ for $t' \in \{t - \tau, \dots, t - 1\}$, the forward variable satisfies

$$\gamma_k(t) = p(\mathbf{c}_t \mid S_k, \mathbf{h}_{t-1}^{(k)}) \sum_{j \neq k} P_{jk} \gamma_j(t-1) \leq 1 \cdot P_{\max,k} \sum_{j=1}^K \gamma_j(t-1) = P_{\max,k} \mathcal{Z}_t^{-1},$$

where $\mathcal{Z}_t$ is the normalisation constant. Applying the recursion backwards for τ steps and using $\sum_j \gamma_j(0) = 1$ with $\gamma_k(0) \leq \pi_k^*$, we obtain the geometric bound $\pi_k^* \cdot P_{\max,k}^\tau$. Since $P_{\max,k} < 1$ (irreducibility from (C3)), the bound decays exponentially with the inter-activation gap τ . This provides a *structural prior on rare emotions* stronger than loss-reweighting alone, which only rescales the gradient without constraining the temporal activation pattern. ■ □

8.2 Monotone Information Gain from Trimodal Fusion

Theorem 8.2: Monotone Information Gain

Let $\mathcal{L}_T$, $\mathcal{L}_{TA}$, and $\mathcal{L}_{TAV}$ denote the maximum marginal log-likelihood achievable using text only, text+audio, and all three modalities respectively. Then

$$\mathcal{L}_T \leq \mathcal{L}_{TA} \leq \mathcal{L}_{TAV},$$

with strict inequalities whenever the additional modality carries emotion information not already encoded in the previous ones.

Proof. The fused context $\mathbf{c}_t = \text{MLP}([\mathbf{z}_t^T \parallel \mathbf{z}_t^A \parallel \mathbf{z}_t^V])$ in ?? is parameterised by the MLP weights. The text-only context $\mathbf{c}_t^T = \text{MLP}([\mathbf{z}_t^T \parallel \mathbf{0} \parallel \mathbf{0}])$ is the special case where audio and video channels are zeroed, achievable by setting the corresponding input-weight columns of the MLP to zero.

Since $\mathcal{L}_{TAV}$ is maximised over *all* MLP parameter values while $\mathcal{L}_T$ is maximised over the strict subset with zeroed audio/video weights, we have $\mathcal{L}_{TAV} \geq \mathcal{L}_T$ by the feasibility argument.

Strict inequality. By the data-processing inequality for mutual information: $I(z_t; \mathbf{c}_t^{TAV}) \geq I(z_t; \mathbf{c}_t^T)$, with equality if and only if $\mathbf{z}_t^A$ and $\mathbf{z}_t^V$ are conditionally independent of z_t given $\mathbf{z}_t^T$. On MELD, prosody (audio) and facial expression (video) provide complementary emotion cues—e.g. sarcastic utterances have neutral text but non-neutral prosody—so $I(z_t; \mathbf{z}_t^A \mid \mathbf{z}_t^T) > 0$ and $I(z_t; \mathbf{z}_t^V \mid \mathbf{z}_t^T, \mathbf{z}_t^A) > 0$ for a non-trivial fraction of utterances, yielding strict inequalities. ■ □

9 Main Theorem

Theorem 9.1: HNMC Theoretical Soundness for MELD

The Hidden Neural Markov Chain with $K = 7$ Mamba experts and trimodal context fusion satisfies all of the following:

- (1) **[Validity]** It defines a proper probability measure over 7-class emotion sequences. (??)
- (2) **[Tractability]** Exact marginal likelihoods are computable in $\mathcal{O}(49T)$ time via the forward algorithm. (??)
- (3) **[Expressiveness]** Each expert universally approximates arbitrary continuous within-regime emotion dynamics. (??)
- (4) **[Identifiability]** Model parameters are recoverable from data up to expert-label permutation under mild rank and ergodicity conditions. (??)
- (5) **[Convergence]** Hybrid EM-backprop training monotonically improves the marginal log-likelihood and converges to a stationary point. (??)
- (6) **[Strict Subsumption]** HNMC strictly contains both classical HMMs and standard Mixture-of-Experts as degenerate special cases. (??)
- (7) **[MELD Structure]** The Markov transition matrix provides automatic geometric regularisation of rare emotion classes and guarantees monotone information gain from trimodal fusion. (????)

Collectively, these results establish that HNMC is a theoretically well-founded architecture with *provable* advantages over its constituent components for conversational multimodal emotion recognition on MELD.

Proof. Each numbered statement is the conclusion of the referenced theorem or proposition, proved individually in ??????????????. The logical dependencies form a directed acyclic graph: ?? underlies ???; ?? underlies ??; ?? underlies all subsequent results. ■ □

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